

JERUSALEM COLLEGE OF ENGINEERING
 (An Autonomous Institution Affiliated to Anna University, Chennai)
 Department of Science and Humanities
 Academic Year: 2024 - 25
QUESTION BANK
JPH2201 ENGINEERING PHYSICS II

*: K1 - REMEMBER, K2 - UNDERSTAND, K3 - APPLY, K4 - ANALYSE, K5 - EVALUATE, K6 -CREATE.

Unit 4 - DIELECTRIC MATERIALS

Course objective - To help them acquire a basic understanding of dielectric materials

Course Outcome - Students will be able to understand the nuances of advanced solid state physics

PART - A

1.	Define dielectric constant. It is the ratio between absolute permittivity of the medium (ϵ) and permittivity of free space (ϵ_0) $\text{dielectric constant} = \frac{\text{Absolute Permittivity}}{\text{Permittivity in free space}}$ $\epsilon_r = \frac{\epsilon}{\epsilon_0}$	CO4	K1
2	Define polarization of dielectric material The process of producing electrical dipoles inside the dielectric by the application an external electrical field is called polarisation in dielectrics. Induced dipole moment (μ) = αE E - Applied electrical field α - Polarizability	CO4	K1
3	Mention four type of polarization mechanisms involved in dielectric. Electronic polarisation Ionic polarisation Orientational polarisation Space - charge polarisation	CO4	K1
4	What is electronic polarization? The induced dipole moment produced in an atom by the application of electric field is known as electronic polarization. It is due to shifting of negatively charged electron and positively charged nucleus of atom in the material by the applied electric field.	CO4	K1
6	What is ionic polarization?	CO4	K1

	Ionic polarisation is due to the displacement of cations (negative ions) and anions (positive ions) in opposite direction due to the application of an electrical field. This occurs in an ionic solid.		
7	What is orientation polarization? When an electrical field is applied on the dielectric medium with polar molecules, the dipoles align themselves in the field direction and thereby increase electric dipole moment. Such a type of contribution to polarisation due to the orientation of permanent dipoles by the applied field is called orientational polarisation.	CO4	K1
8	What is space charge polarization? In some materials containing two or more phases, the application of an electrical field causes the accumulation of charges at the interfaces between the phases or at the electrodes. As result of this, polarisation is produced. This type of polarisation is known as space charge polarisation	CO4	K1
9	What is meant by local field in a dielectric? When a dielectric is kept in an external electric field (E), two fields are exerted due to (i) external field and (ii) dipole moment created. These long range coulomb forces which are created due to local field in dielectric. $E_{int} = E + \frac{P}{3\epsilon_0}$	CO4	K1
10	Define dielectric loss When a dielectric material is subjected to an A.C voltage, the electrical energy is absorbed by the material and is dissipated in the form of heat. This dissipation of energy is called dielectric loss.	CO4	K1
11	Define dielectric breakdown and dielectric strength. Whenever the electrical field strength applied to a dielectric exceeds a critical value, very large current flows through it. The dielectric loses its insulating property and becomes conducting. This phenomenon is known as dielectric breakdown. The electrical field strength at which dielectric breakdown occurs is known as dielectric strength	CO4	K1
12	What are the different types of breakdown mechanisms? Intrinsic breakdown and avalanche breakdown Thermal breakdown Discharge breakdown Chemical and Electrochemical breakdown Defect breakdown	CO4	K1
13	Define intrinsic breakdown. For a dielectric, the charge displacement increases with increasing electrical field strength. Beyond a critical value of electrical field strength, there is an electrical breakdown due to physical deterioration in the dielectric material.	CO4	K1
14	What is thermal breakdown?	CO4	K1

	When an electrical field is applied to a some amount of dielectric material, heat is produced. This heat must be dissipated from the material. Due to excess of heat, the temperature inside the dielectric increases and may produce local melting in the dielectric material. This type of breakdown is known as thermal breakdown				
15	What is chemical and electro chemical breakdown? Electrochemical breakdown is similar to thermal breakdown. When the temperature of a dielectric material increases, mobility of ions increases and hence the electrochemical reaction may take place. This leads to leakage current and energy loss in the material and finally dielectric breakdown occurs.			CO4	K1
16	What is discharge break down? Discharge breakdown occurs when a dielectric contains occluded gas bubbles. When this type of dielectric is subjected to electrical field, the gases present in the material will easily ionise and thus produce large ionisation current. The gaseous ions bombard the solid dielectric. This causes electrical deterioration and leads to dielectric breakdown.			CO4	K1
17	What is defect breakdown? The surface of the dielectric material may have defects such as cracks, porosity and blow holes. Impurities like dust or moisture may collect at these discontinuities (defects). This will lead to a breakdown in a dielectric material.			CO4	K1
18	Compare active and passive dielectrics.			CO4	K2
	S. No	Active dielectrics	Passive dielectrics		
	1.	Dielectrics which can easily adapt itself to store the electrical energy in it is called active dielectrics.	Dielectric which restricts the flow of electrical energy in it are called passive dielectrics.		
	2.	Examples: Piezo electrics, Ferro electrics, Pyro electrics	Examples: Glass, Mica and Plastic		
	3.	It is used in the production of ultrasonic	It is used in the production of sheets, pipes etc.		
19	What are the differences between polar and non-polar molecules?			CO4	K1
	S. No	Polar	Non-Polar		
	1.	These molecules have permanent dipole moments even in the absence of an applied field.	These molecules do not have permanent dipole moments.		
		The polarization of polar molecules is highly temperature dependent	The polarization of this kind of molecules is independent of temperature.		
		These molecules do not have symmetrical structure and they do not have center of symmetry.	These molecules have symmetrical structure and they have center of symmetry. .		

		Examples: CHCl_3 , H_2O , HCL	Examples: CCl_4 , CO_2 , H_2		
20	What are the uses of dielectric materials? It is used as a dielectric medium in capacitor It is used as insulating medium in transformer It is used in industries and dielectric heating.			CO4	K1

PART B

1	Explain the different types of polarization mechanisms involved in a dielectric material.	CO4	K2
2	Derive an expression for electronic and ionic polarisabilities.	CO4	K3
3	Discuss in detail the various dielectric breakdown mechanisms.	CO4	K1
4	Define Internal field obtained the expression for internal field using Lorentz method and hence deduce the Clausius-Mosotti equation.	CO4	K3
5	Discuss the applications of dielectric materials in capacitors and transformers.	CO4	K1
6	Explain dielectric loss and find an expression for dielectric power loss.	CO4	K2

Unit 5 – NANOELECTRONIC MATERIALS

Course objective – To introduce the basic principles in nano electronic devices To help them acquire a basic understanding of dielectric materials

Course Outcome – Students will be able to recognize the uses of Nano electronic materials

PART - A

1.	Define Nano materials. The material which is having particle size of 1nm (or 10^{-9} m) is called Nano materials.	CO5	K1
2	Define Density of States. Electron density is the number of electrons per unit volume in a material. It is determined by using density of states.	CO5	K1
3	Define Fermi energy. It is defined as the highest energy level occupied by the electron at 0K in metal	CO5	K1
4	What is Quantum Confinement? The Process of reduction of the size of the solid such that the energy levels inside become discrete.	CO5	K1
6	What is Quantum structure? When a bulk material is reduced in its size, at least one of its dimension, in the order of few nanometres, then the structure is known as quantum structure.	CO5	K1
7	What is Single Electron tunnelling? The phenomena of penetration of charge carriers directly through the potential barrier, instead of climbing over it is called tunnelling.	CO5	K1
8	What is single electron transistor? Mention its advantage and applications? A transistor made from a quantum dot that controls the current from source to drain one electron at a time is called single electron transistor. Advantages Information transferring velocity is high No wire is needed between the memory chips.	CO5	K1

	Applications It is used in sensor technology and digital electronic circuits. Used in single electron spectroscopy		
9	What are the limitations of single electron transistor? Size of the quantum dot should be smaller than 10nm in order to operate it is the room temperature. Very hard to fabricate by traditional methods.	CO5	K1

PART B

1	Explain the electron density in bulk material and size dependence of Fermi energy.	CO5	K3
2	Explain quantum confinement and quantum structures in nano material.	CO5	K3
3	Discuss density of states in quantum well, quantum wire, and quantum dot structure.	CO5	K3
4	Describe construction and working of single electron transistor.	CO5	K2
5	Describe single electron phenomena and single electron transistor.	CO5	K1